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An Evidence-Based Component Attribution Framework

Presented by
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Dedication

This work is dedicated to my parents,

to my dear sister and brother,

to my entire family,

and to the friends who accompanied and supported me throughout my journey, from elementary school to graduation.

Acknowledgements

First and foremost, I thank God for granting me the strength, patience, and perseverance necessary to complete this work and reach this important stage of my academic journey.

I express my deepest gratitude to my parents and my entire family. Their constant encouragement, sacrifices, trust, and unconditional support have shaped the person I am today. This achievement would not have been possible without the values they taught me and the confidence they placed in me.

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Abstract

Software Bills of Materials are increasingly used to support software supply-chain transparency and vulnerability management. However, a syntactically valid SBOM is not automatically trustworthy, particularly when it is reconstructed from a compiled Android application without source code or dependency manifests. Compilation, optimization, obfuscation, artifact-family aliases, reference-database gaps, and version similarity can weaken or contradict the evidence supporting a component claim.

This thesis studies trustworthy SBOM recovery as an evidence-based software component attribution problem. It organizes existing Android third-party library and binary component-recovery research into taxonomies of techniques, evidence sources, transformation threats, and evaluation practices. Building on the resulting research gaps, it proposes a conceptual framework in which each recovered claim records component identity, version or version set, evidence provenance, evidence quality, transformation risk, confidence, ambiguity, validation class, and permitted downstream security interpretation.

The implemented APK-to-SBOM PFE system is used as a bounded case study. Its controlled, clean and partial source-validated, source-known upper-bound, experiment-only, report-only, and boundary evidence demonstrates why library identity, exact version, reference coverage, matcher recall, and vulnerability authority must remain separate. A case-study evaluation confirms that the framework preserves these evidence lanes and blocks authority increases when apparently beneficial changes fail ambiguity, control, or promotion gates. The proposed framework therefore favors auditable partial knowledge and explicit abstention over unsupported certainty, and defines a research agenda for calibrated evidence fusion, provenance-aware SBOMs, realistic benchmarks, and native or dynamically loaded component recovery.

Keywords — software bill of materials, software supply chain, Android, component attribution, provenance, evidence fusion, trustworthiness, uncertainty, vulnerability management.

Résumé

Les nomenclatures logicielles, ou SBOM, sont de plus en plus utilisées pour renforcer la transparence de la chaîne d'approvisionnement logicielle et la gestion des vulnérabilités. Toutefois, une SBOM syntaxiquement valide n'est pas automatiquement digne de confiance, en particulier lorsqu'elle est reconstruite à partir d'une application Android compilée, sans accès au code source ni aux manifestes de dépendances. La compilation, l'optimisation, l'obfuscation, les alias de familles d'artefacts, les lacunes des bases de références et la similarité entre versions peuvent affaiblir ou contredire les preuves qui soutiennent une déclaration de composant.

Ce mémoire étudie la reconstruction fiable de SBOM comme un problème d'attribution de composants logiciels fondé sur les preuves. Il organise les travaux existants sur la détection de bibliothèques tierces Android et la récupération de composants binaires selon des taxonomies de techniques, de sources de preuves, de menaces de transformation et de pratiques d'évaluation. À partir des lacunes ainsi mises en évidence, il propose un cadre conceptuel dans lequel chaque déclaration reconstruite conserve l'identité du composant, la version ou l'ensemble de versions, la provenance des preuves, leur qualité, le risque de transformation, le niveau de confiance, l'ambiguïté, la classe de validation et l'interprétation de sécurité autorisée en aval.

Le système PFE d'analyse APK-vers-SBOM sert d'étude de cas délimitée. Ses preuves contrôlées, propres et partielles validées par les sources, ses bornes supérieures connues par les sources, ainsi que ses résultats expérimentaux, indicatifs et limites, montrent pourquoi l'identité de bibliothèque, la version exacte, la couverture des références, le rappel du moteur de correspondance et l'autorité des vulnérabilités doivent rester séparés. Le cadre est évalué par l'étude de cas : il préserve ces catégories de preuves et empêche toute augmentation d'autorité lorsqu'une modification apparemment bénéfique échoue aux critères d'ambiguïté, de contrôle ou de promotion. Le cadre proposé privilégie donc une connaissance partielle auditable et l'abstention explicite plutôt qu'une certitude non étayée, et définit un programme de recherche pour la fusion calibrée des preuves, les SBOM sensibles à la provenance, les bancs d'essai réalistes et la récupération de composants natifs ou chargés dynamiquement.

Mots-clés — nomenclature logicielle, chaîne d'approvisionnement logicielle, Android, attribution de composants, provenance, fusion de preuves, confiance, incertitude, gestion des vulnérabilités.

ملخص

تُستعمل قوائم مكونات البرمجيات، أو SBOM، بشكل متزايد لدعم شفافية سلسلة توريد البرمجيات وإدارة الثغرات. غير أن وجود SBOM صحيحة من الناحية التركيبية لا يعني بالضرورة أنها موثوقة، خاصة عندما يعاد بناؤها من تطبيق أندرويد مترجم دون الوصول إلى الشفرة المصدرية أو ملفات بيان الاعتماديات. فقد تؤدي الترجمة، والتحسين، والتعمية، وأسماء عائلات القطع البرمجية، ونقص قواعد المراجع، والتشابه بين الإصدارات إلى إضعاف الأدلة التي تدعم ادعاء وجود مكون معين أو إلى مناقضتها.

يدرس هذا العمل استرجاع SBOM الموثوقة بوصفه مسألة إسناد مكونات برمجية قائمة على الأدلة. وينظم الأبحاث السابقة حول كشف المكتبات الخارجية في أندرويد واسترجاع المكونات الثنائية ضمن تصنيفات للتقنيات، ومصادر الأدلة، ومخاطر التحويل، وممارسات التقييم. وبناء على الفجوات الناتجة، يقترح إطارا مفهوميا يسجل لكل ادعاء مسترجع هوية المكون، والإصدار أو مجموعة الإصدارات، ومصدر الأدلة، وجودتها، ومخاطر التحويل، ومستوى الثقة، والغموض، وفئة التحقق، والتفسير الأمني المسموح به لاحقا.

يُستخدم نظام مشروع نهاية الدراسة لتحليل APK وإنتاج SBOM كدراسة حالة محدودة. وتبين أدلته المضبوطة، والنتائج النظيفة والجزئية المحققة بالمصدر، والحدود العليا المعروفة بالمصدر، والنتائج التجريبية أو الإرشادية أو الحدية، ضرورة الفصل بين هوية المكتبة، والإصدار الدقيق، وتغطية المراجع، واسترجاع المطابق، وسلطة ادعاءات الثغرات. ويؤكد تقييم دراسة الحالة أن الإطار يحافظ على مسارات الأدلة هذه ويمنع رفع سلطة الادعاء عندما تفشل التغييرات التي تبدو مفيدة في اختبارات الغموض أو الضبط أو الترقية. لذلك يفضل الإطار المقترح معرفة جزئية قابلة للتدقيق وامتناعا صريحا عن الحكم بدلا من يقين غير مدعوم، كما يحدد برنامجا بحثيا لدمج الأدلة بصورة معايرة، ولبناء SBOM واعية بالمصدر، ولاختبارات واقعية، ولاسترجاع المكونات الأصلية أو المحملة ديناميكيا.

الكلمات المفتاحية — قائمة مكونات البرمجيات، سلسلة توريد البرمجيات، أندرويد، إسناد المكونات، مصدر الأدلة، دمج الأدلة، الثقة، عدم اليقين، إدارة الثغرات.

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List of Acronyms

Acronym	Definition
APK	Android Package Kit
API	Application Programming Interface
CVE	Common Vulnerabilities and Exposures
DEX	Dalvik Executable
ML	Machine Learning
NVD	National Vulnerability Database
OSV	Open Source Vulnerabilities
PURL	Package URL
R8	Android code shrinker and optimizer
SBOM	Software Bill of Materials
SCA	Software Composition Analysis
SPDX	Software Package Data Exchange
TPL	Third-Party Library

Part I

General Introduction

Chapter 1

General Introduction

1.1 Context

Software supply chains depend on large ecosystems of reusable components. SBOMs improve transparency by recording these components, but many audit scenarios provide only compiled software artifacts. In such cases, an SBOM is reconstructed from indirect and transformed evidence rather than generated from authoritative dependency metadata.

1.2 Problem Statement

A recovered component name or version is not trustworthy merely because a tool emitted it. Trust depends on the origin, independence, transformation resilience, version specificity, and validation status of its evidence. Existing Android detection research provides strong technical signals, but lacks a unified model connecting those signals to claim provenance, uncertainty, and permitted vulnerability interpretation.

1.3 Motivation

Software transparency is most useful when an inventory can be inspected, challenged, and updated. Build-generated dependency metadata can often provide direct package coordinates and resolved versions, but auditors, incident responders, application stores, and downstream consumers may receive only a compiled artifact. In that setting, reconstructing an SBOM is necessary, yet the result is an inference from residual binary evidence rather than a direct export of the original build graph.

This distinction matters for vulnerability management. A package name may be sufficient to guide an investigation while remaining insufficient for an exact-version or affected-range decision. Conversely, an exact-looking version can be unsafe when the true release is absent from the reference base, when adjacent releases are feature-equivalent, or when optimization has removed the few discriminating features. If these conditions are hidden, a syntactically valid SBOM can encourage unsupported remediation priorities or false assurance.

The research reviewed in this thesis provides many useful signals for component recovery, including names, class and method structure, opcodes, literals, dependency relations,

graphs, metadata, and learned representations. However, technical evidence and evaluation results answer different questions. Library identity, exact-version attribution, reference coverage, matcher recall, source validation, and vulnerability authority must not be collapsed into one confidence value or success rate.

The academic motivation is therefore to formalize how recovered evidence becomes an auditable claim. In this thesis, a reconstructed claim is *trustworthy* when its identity and version strength, supporting and opposing evidence, provenance, transformation exposure, unresolved uncertainty, validation context, and permitted downstream use are visible and mutually consistent. This definition allows partial knowledge and abstention to remain useful outcomes rather than forcing every analysis into an unsupported exact answer.

1.4 Research Questions

The central research question is:

> *How can heterogeneous evidence extracted from compiled and obfuscated Android applications be combined to produce trustworthy SBOM component and version claims with explicit provenance, confidence, uncertainty, and validation boundaries?*

This question is developed through six supporting questions:

1. Which evidence sources are available for recovering software components from compiled Android applications?
2. How have existing Android library-detection and binary component-recovery approaches combined these sources?
3. Which factors determine evidence strength for identity and exact-version attribution?
4. How should conflicting, missing, transformed, or source-dependent evidence be represented?
5. What information must accompany a reconstructed SBOM claim for it to be auditable and useful in vulnerability management?
6. How can such a framework be evaluated without overstating accuracy?

1.5 Objectives and Contributions

The objective is to connect Android binary component-recovery techniques to a clear model of claim trust. The thesis reviews and classifies prior work, identifies the conditions that strengthen or weaken identity and version evidence, formalizes an evidence-based attribution framework, illustrates the framework through selected PFE observations, and derives directions for further research.

The resulting contributions are:

1. a recovery-oriented taxonomy of techniques, evidence sources, transformation threats, and evaluation practices;

2. operational trust requirements based on provenance, evidence visibility, identity/version separation, uncertainty, validation context, vulnerability authority, and explicit denominators;
3. a structured recovered-claim model covering identity, version, relationships, boundary observations, and downstream interpretation;
4. an evidence hierarchy and fusion policy with conflict handling, confidence caps, bounded ambiguity, and abstention;
5. a validation-lane and vulnerability-authority model that prevents a security interpretation from exceeding its component evidence; and
6. a PFE-informed framework case study and research agenda for broader calibration, validation, interoperability, and component domains.

These are research and framework contributions. The associated APK-to-SBOM software implementation and its detailed engineering evaluation belong to the PFE; this thesis uses selected project evidence only to motivate and examine the proposed concepts.

1.6 Scope

The primary domain is static reconstruction of software-component claims from compiled Android APKs. The analysis focuses on DEX-observable JVM/Maven components, versioned reference artifacts, and the implications of recovered identity and version claims for SBOM production and vulnerability management. Android is used as a demanding case because compilation, shrinking, optimization, obfuscation, multidex packaging, artifact variants, and incomplete references can all change the evidence available after build time.

The thesis covers:

- structured review and taxonomy construction for Android third-party library and binary component-recovery research;
- evidence quality, provenance, transformation exposure, ambiguity, and validation boundaries for recovered claims;
- claim classes for exact, bounded, identity-only, report-only, abstained, and boundary outcomes;
- constraints on component-to-vulnerability interpretation; and
- selected PFE observations used as case-study evidence.

The thesis does not propose a second APK detector, repeat the complete PFE implementation description, or present the PFE evaluation as independent master experiments. It does not claim universal probabilistic calibration, complete dependency recovery, or general native package/version attribution. Native shared objects, runtime-loaded code, encrypted payloads, commercial protectors, and other unsupported component domains are represented as boundary categories and future-research problems. The framework is therefore conceptual and evidence-informed: it defines how claim authority should be bounded, but it is not a formal proof that every relevant component can be observed or identified.

1.7 Thesis Organization

The remainder of the thesis is organized as follows.

Chapter 2 establishes the conceptual and methodological foundation. It distinguishes authoritative and reconstructed SBOMs, defines evidence, provenance, uncertainty, and software-component attribution, describes the Android compiled-artifact context, and records the structured literature-review method. Appendix A contains the detailed review protocol, selection rules, extraction schema, quality assessment, and audit controls.

Chapter 3 applies that method to the selected literature. It constructs taxonomies of recovery techniques, evidence sources, transformation threats, and evaluation practices, compares representative approaches without treating incompatible reported scores as one leaderboard, and derives the research gap addressed by the thesis.

Chapter 4 presents the principal conceptual contribution: an evidence-based trust framework for reconstructed SBOM claims. It defines the recovered-claim schema, evidence hierarchy, quality dimensions, fusion and conflict policy, claim classes, validation lanes, vulnerability-authority rules, claim lifecycle, and bounded PFE case study.

Chapter 5 answers the research questions, summarizes the contributions and limitations, and develops future directions including empirical calibration, independent validation, provenance-aware exchange, native and dynamic attribution, human-centered evaluation, and evidence knowledge graphs.

Part II

Background and Research Methodology

Chapter 2

Background and Research Methodology

2.1 Introduction

This chapter establishes the concepts and review method used to reason about trustworthy recovery of software components from compiled Android applications. The thesis treats a recovered SBOM as a set of evidence-backed claims rather than as a simple list of names and versions. This distinction requires a common vocabulary for component identity, provenance, uncertainty, and transformation risk, followed by a transparent method for selecting and comparing prior research.

2.2 Software Supply Chains and SBOMs

Modern applications assemble code, resources, native binaries, generated artifacts, and transitive dependencies from many independently maintained projects. An SBOM records these components so that operators can reason about inventory, licensing, provenance, and known vulnerabilities. Standards define how such information is represented, but a representation format does not establish that the recorded component is actually present or that its exact version is known. This distinction is consistent with supply-chain guidance that treats component transparency as one control within a broader risk management process, and with empirical SBOM research showing persistent generation, consistency, and consumption challenges [1, 2, 3].

The framework considers both SPDX and CycloneDX, while using Package URL as a standard package-identity model [4, 5, 6]. These standards therefore define the target vocabulary and exchange structures, while the attribution framework developed in this thesis defines the evidential conditions under which individual fields may be populated.

2.2.1 Authoritative and Reconstructed SBOMs

An authoritative build-time SBOM can use dependency resolution metadata, lockfiles, build graphs, and package-manager records. By contrast, an artifact-reconstructed SBOM begins after those authoritative records have been removed or made unavailable. It must

infer a component from observable residue in an APK, compare that residue with references, and preserve the uncertainty introduced by compilation and incomplete coverage.

The distinction is central to this thesis. A reconstructed component identity may be credible even when its exact version is unresolved. Similarly, a syntactically valid CycloneDX or SPDX document may contain semantically weak claims. Trustworthiness is consequently a property of each claim and its evidence trail, not of the file format alone.

2.3 Trust, Evidence, Provenance, and Uncertainty

2.3.1 Evidence and Claims

An evidence item is an observable fact used to support or challenge a component claim. Examples include a package fragment, an opcode fingerprint, a class-dependency relation, a resource marker, a native-library name, or a source-validated expected dependency. A claim is the interpretation built from one or more such items, for example that a library family is present or that a particular version is the best-supported candidate.

2.3.2 Provenance and Independence

Provenance records where evidence came from, how it was extracted, and which transformations may have affected it. Two matching fields are not independent when they are derived from the same underlying artifact feature. Confidence must therefore not be increased merely by counting correlated signals as separate confirmations. The framework adopts the W3C PROV distinction between entities, activities, and agents as a general provenance vocabulary, while specializing it for APK observations, reference profiles, validation artifacts, and analysis steps [7].

2.3.3 Uncertainty and Abstention

Uncertainty can arise from missing evidence, conflicting evidence, incomplete reference coverage, adjacent versions with near-identical code, partial library inclusion, or destructive optimization. The framework allows ambiguity sets and abstention because an explicit unresolved result is more auditable than a forced exact version. Reject-option research provides the general decision-theoretic precedent for withholding a classification when the expected error cost is too high [8]. This policy also limits downstream vulnerability authority: a vulnerability finding cannot be stronger than the identity and version evidence on which it depends.

2.4 Android Compiled-Artifact Context

Android compilation and R8 transformations directly affect the stability and interpretability of recovered evidence [9].

Java or Kotlin source is compiled and transformed into DEX bytecode, while resources and manifests are packaged into an APK. Dependency boundaries that were explicit during the build may disappear when classes are merged into one or more DEX files. R8

can rename identifiers, remove unused code, rewrite control flow, merge classes, inline methods, and change call sites. Packaging may also introduce multidex layouts, shaded or relocated code, native libraries, generated classes, and dynamically loaded components.

These transformations do not affect all evidence equally. Names and package trees are inexpensive but fragile. Opcode, control-flow, call, and dependency structures can survive renaming, yet they are still altered by shrinking and optimization. Resources and metadata can provide independent support, but they may identify only a family rather than an exact version. The state-of-the-art taxonomy in chapter 3 therefore classifies both evidence sources and the transformations that threaten them.

2.5 Software Component Attribution

2.5.1 Identity, Version, and Security Authority

Component attribution contains at least three distinct decisions:

1. **identity attribution**: whether a component family or coordinate is present;
2. **version attribution**: which exact version or bounded version set is supported; and
3. **security interpretation**: which vulnerability statements are permitted by the preceding evidence.

A method that performs well at library-family detection does not automatically provide exact-version authority. Likewise, a vulnerability database match is not confirmation that the affected version is present. Keeping these decisions separate prevents a plausible identity match from being silently promoted into an exact and security-critical statement.

2.5.2 Reference Dependence

Many attribution methods compare an application with profiles extracted from known library releases. Such a method can recognize only what its reference corpus represents. Failure may therefore result from the matcher, the extractor, or a missing reference rather than from absence of the component. Reference coverage is treated as a first-class validity condition throughout the thesis.

2.6 Evidence Sources in APKs

The review extracts evidence into recurring families:

- lexical and namespace evidence, including package and class names;
- bytecode and semantic evidence, including opcodes, API references, literals, and method descriptors;
- structural evidence, including control-flow, call, inheritance, and class-dependency graphs;

- resource and packaging evidence, including manifests, resource names, assets, and native-library files;
- external reference and metadata evidence, including library profiles, Maven coordinates, dependency relations, and release chronology;
- learned representations derived from bytecode or structural features;
- validation evidence, including controlled builds, source manifests, and source-based observations.

No family is assumed to be universally authoritative. The framework evaluates its origin, transformation resilience, version specificity, independence, coverage, and validation status.

2.7 Structured Literature Review Methodology

The literature study follows a frozen structured mapping protocol adapted from software-engineering systematic-review guidance [10]. It is not presented as a completed exhaustive systematic literature review because the initial corpus was assembled iteratively before the protocol freeze. Instead, the protocol makes the existing corpus auditable and governs all future search refreshes, screening, extraction, and synthesis.

2.7.1 Review Questions and Search Strategy

The review asks which evidence sources and attribution techniques are used, which Android transformations threaten them, how identity and version results are evaluated, how uncertainty and provenance are represented, and which gaps prevent trustworthy SBOM claims. Searches combine four concept blocks: Android artifacts; third-party components or SBOMs; detection or attribution tasks; and obfuscation, optimization, version, or vulnerability terms.

Primary search sources include ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, USENIX, and arXiv. Google Scholar is restricted to discovery and citation chaining. The date scope begins in 2015 and continues through the recorded search date. The canonical query and the recording requirements for future database-specific adaptations are preserved in chapter A. Because the seed corpus predates the protocol freeze, the original database-specific query executions and result counts cannot be reconstructed retrospectively.

2.7.2 Eligibility and Screening

Eligible studies propose or evaluate Android binary component attribution, version identification, relevant benchmarks, empirical assessments, or SBOM-oriented recovery frameworks. Studies based only on authoritative source manifests, unrelated malware classification, product pages, duplicates, and records without extractable methodology are excluded.

Records are deduplicated and screened by title and abstract. Full-text eligibility assessment and structured extraction are completed for Tier A; records without a verified local full text remain Tier B pending that stage. Related conference and journal publications

are treated as one study family unless the later publication contributes distinct methods, datasets, or results.

2.7.3 Evidence Tiers

The frozen seed corpus contains 20 publication units. Fifteen have locally inspected full texts and form *Tier A*. Five are *Tier B* supplementary records whose metadata or structured notes are available but whose local full texts have not yet been verified. Tier B records may inform search coverage and provisional gap mapping, but they cannot independently support precise quantitative conclusions. Standards and official platform documentation form contextual *Tier C* and are not counted as primary studies.

2.8 Analysis Dimensions

Each included study is extracted across common dimensions: task granularity, input assumptions, segmentation, evidence families, matching or learning method, transformation coverage, ground truth, dataset composition, metrics, confidence and ambiguity behavior, reference dependence, artifact availability, SBOM output, vulnerability mapping, and stated limitations.

Tier A studies are additionally assessed against eight reporting-quality questions covering objective clarity, reproducibility, ground truth, metric definition, transformation conditions, identity/version separation, artifact availability, and threats to validity. Quality affects the weight of a study in synthesis but does not automatically erase historically important work.

Synthesis is narrative and taxonomy-based. Metrics are compared only when their task unit, dataset, ground truth, and transformation conditions are compatible. Original-paper results remain separate from later independent benchmarks, and library-level, class-level, and version-level results are never merged into a single ranking.

2.9 Threats to the Review

2.9.1 Search and Selection Bias

The pre-protocol corpus was built iteratively and may overrepresent well-known Android security venues or studies connected through citation chains. The frozen search blocks, multiple publisher databases, and explicit refresh log reduce but do not eliminate this risk. The original database-specific queries, raw result counts, and deduplicated flow were not retained before the freeze, so the protocol cannot retroactively establish an exhaustive or PRISMA-style selection history for the seed corpus. Those fields are mandatory for every future refresh.

2.9.2 Availability and Recency

Several recent or emerging studies currently exist only as supplementary records. Treating them as Tier B prevents incomplete metadata from acquiring the same authority as

inspected full text. TPL-Benchmark and UniBOM were promoted after local full-text, bibliographic, extraction, and quality checks; the remaining classifications may change only after the same checks, and all such changes must be logged.

2.9.3 Heterogeneous Evaluation

Reported accuracy values often refer to different task definitions, datasets, library versions, obfuscators, and ground-truth procedures. Direct numerical ranking can therefore be misleading. The review preserves these conditions and prefers qualified comparison over pooled scores.

2.9.4 Reviewer and Extraction Bias

The current review is primarily maintained by one researcher. A fixed extraction schema, explicit `NR` values for unreported information, source citations, and retained structured records provide an audit trail. A later independent screening sample would further strengthen reliability.

2.10 Conclusion

This chapter separated a valid SBOM representation from the evidential strength of its contents and defined component attribution as a sequence of identity, version, and security-authority decisions. It also froze a structured mapping protocol that prevents unavailable full texts, incompatible metrics, and correlated signals from being silently promoted into strong claims. The next chapter applies these rules to build the state-of-the-art taxonomies that motivate the proposed trust framework.

Part III

State of the Art and Taxonomy

Chapter 3

State of the Art and Taxonomy

3.1 Introduction

This chapter applies the structured mapping protocol defined in chapter 2 and chapter A to the Android third-party library detection literature. Its purpose is not to produce a numerical leaderboard. The reviewed studies use different datasets, task units, obfuscation settings, and ground-truth procedures, so a direct ranking would hide the very uncertainty that this thesis aims to expose. Instead, the chapter organizes prior work by technique, evidence source, transformation threat, and evaluation practice.

The analysis is centered on the 15 Tier A studies whose full texts are available locally. Tier B records are useful for search coverage and for identifying emerging directions, especially SBOM-oriented recovery and recent benchmarking, but they are not used here as independent authority for precise quantitative claims. This boundary keeps the review consistent with the protocol and prevents unverified metadata from receiving the same weight as inspected studies. Fields not reported by a Tier A publication are treated as not reported rather than inferred, and original-paper claims are kept separate from later cross-tool observations.

The chapter uses the term *component recovery* broadly. A study may recover a library family, a package cluster, a class set, an exact version, a vulnerability-related version list, or only a presence signal. For trustworthy SBOM reconstruction, these outputs are not equivalent. A method that detects the presence of a library family does not necessarily justify an exact Maven version, and an exact-version candidate does not automatically justify a CVE claim unless the evidence, reference coverage, and validation context are visible.

3.2 Taxonomy of Component-Recovery Techniques

The reviewed approaches can be grouped into seven recurring technique families. The categories are not mutually exclusive: most mature tools combine several families, for example by using a fast coarse filter followed by a slower structural match. The taxonomy therefore describes the role each family plays in a recovered component claim.

Table 3.1: Technique taxonomy for Android component recovery.

Technique family	Typical evidence	Trust role	Tier A examples
Namespace and metadata heuristics	Package trees, class names, library naming conventions	Cheap candidate generation, but weak authority when names are renamed, flattened, relocated, or shaded.	Assessed in [11, 12]
Profile, hash, and signature matching	Method descriptors, Merkle trees, fuzzy hashes, class or method signatures	Can support identity and, when references are complete, version claims. Its authority is bounded by reference coverage and by the stability of the hashed features under optimization.	LibScout [13], ORLIS [14], LibID [15]
Structural graph segmentation	Call, inheritance, containment, and class-dependency relations	Improves module boundary recovery when package structure is unreliable. Supports stronger identity evidence, but does not by itself solve exact version attribution.	LibD [16], LibPecker [17], assessment studies [11, 12]
Coarse-to-fine matching	Fast filtering followed by detailed opcode, CFG, or structural comparison	Balances scalability and precision. Trust depends on both stages: a missed coarse candidate can cap recall, while an over-permissive fine stage can inflate false positives.	ORLIS [14], ATVHUNTER [18], LibScan [19]
Version-discriminating semantic features	Opcode sequences, basic-block features, call chains, field-operation patterns, literals, and class summaries	Provides the main route from library identity to exact-version authority. Adjacent versions and optimized builds remain difficult because many versions share most code.	ATVHUNTER [18], LibScan [19], LibHunter [20], SAD [21]
Learned representations	Smali token sequences, structural vocabularies, neural embeddings	Can learn obfuscation-resilient presence signals, but the reviewed Tier A learning approach does not provide exact-version or boundary authority on its own.	LibAttention [22]

Technique family	Typical evidence	Trust role	Tier A examples
Empirical assessment and benchmarking	Shared datasets, patched tools, source-known applications, evaluation criteria	Does not directly recover components, but defines which claims are comparable and exposes tool failure modes, missing denominators, and reproducibility limits.	Tool assessments [11, 12], Ramirez thesis [23]

The movement across these families is historically important. Early methods made library profiles and package structure usable at scale, but later studies increasingly treat package names as unreliable evidence. Graph-based and version-aware methods improve the situation by using relations, opcode content, and class behavior. The newest verified Tier A work adds learned representations and finer class-level summaries. Even so, the taxonomy shows that standard SBOM emission remains outside the main output model of the verified Android TPL literature: most studies output names, clusters, versions, class mappings, or vulnerability-linked lists rather than a provenance-rich CycloneDX or SPDX document.

3.3 Taxonomy of Evidence Sources

An evidence source is useful for trustworthy SBOM recovery only when its origin, stability, version specificity, and provenance are understood. The same signal may be strong for identity and weak for exact version. Conversely, a highly version-specific literal may be too fragile to survive resource removal or string encryption. Table 3.2 summarizes the recurring evidence families extracted from the Tier A studies.

Table 3.2: Evidence-source taxonomy for reconstructed Android component claims.

Evidence family	Examples	Typical stability	Trust implication
Names and namespaces	Package prefixes, class names, manifest names	Weak under renaming, package flattening, relocation, and shading	Useful for fast hints and human inspection, but insufficient for exact SBOM claims without corroboration.
Descriptors and API surfaces	Method signatures, fuzzy type descriptors, Android API usage	More stable than names under identifier renaming; weaker under dead-code removal and inlining	Supports identity and coarse version discrimination when combined with reference profiles.

Evidence family	Examples	Typical stability	Trust implication
Opcode and instruction sequences	Basic-block opcodes, method opcode chains, call-chain opcodes	Resilient to naming changes; threatened by optimization, inlining, and control-flow rewriting	Provides version-discriminating evidence, especially when compared against known library releases.
Control-flow and dependency structures	CFGs, call graphs, inheritance graphs, class-dependency graphs	Stable against simple renaming and often against package restructuring; affected by optimization and partial inclusion	Improves module boundary recovery and reduces dependence on package layout.
Literals, constants, and resources	Strings, constants, resource identifiers, assets	Potentially version-specific, but fragile under stripping, string encryption, and resource rewriting	Best used as tie-breakers or independent corroboration, not as sole authority.
Reference corpus and release metadata	Library profiles, Maven coordinates, release chronology, known vulnerable version ranges	External to the APK; stable only if the corpus is complete and correctly constructed	Defines the ceiling for recognizable identities and versions. Missing references must not be interpreted as component absence.
Learned representations	Transformer embeddings over Smali tokens, structural vocabularies	Can absorb complex obfuscation patterns, but may be hard to audit feature by feature	Promising for presence screening and robustness, but provenance and exact version authority require additional explanation.
Validation evidence	Controlled builds, source manifests, F-Droid dependency data, source-known ground truth	Strong within the validation context; not automatically generalizable	Can upgrade a claim's evaluation status, but must remain separate from APK-only inference.

The trust implication of this taxonomy is that evidence fusion should prefer independent and complementary evidence over repeated variants of the same feature. For example, several hash matches derived from the same method body are less independent than a method-level match corroborated by a class-dependency relation and a version-specific literal. The framework in chapter 4 builds on this distinction by treating independence, provenance, and reference coverage as first-class quality dimensions.

3.4 Taxonomy of Transformation Threats

Android component recovery is difficult because compilation and packaging can remove the very boundaries that build-time dependency tools rely on. R8 and related obfuscators

may rename identifiers, flatten packages, remove unused code, rewrite control flow, and inline methods [9]. The reviewed literature shows that no single evidence family remains stable under all transformations.

Table 3.3: Transformation-threat taxonomy for Android component recovery.

Threat	**Evidence most affected**	**More resilient evidence**	**Trust consequence**
Identifier renaming	Class, method, field, and package names	Fuzzy descriptors, opcode features, structural relations	Name matches cannot be treated as independent authority unless corroborated.
Package flattening and relocation	Package-subtree segmentation and namespace heuristics	Class-dependency graphs, call relations, profile matching independent of package roots	Package-based module boundaries become unreliable; graph segmentation becomes more important.
Control-flow rewriting	CFG shape, branch structure, fuzzy CFG hashes	Higher-level dependency relations, method descriptors, class summaries	CFG-only methods may overstate robustness when control-flow transformations are not in the evaluation.
Dead-code removal and shrinking	Presence of optional classes, methods, API calls, literals	Partial-match scoring, class-level confidence, source-known validation	Absence of a feature may indicate transformation rather than absence of the library.
Method inlining and outlining	Method boundaries, call-chain evidence, version-specific method bodies	Multi-scale evidence, class-level summaries, literals, residual dependency structure	Exact-version authority must be capped when inlining changes the features used for version discrimination.
Multidex and split packaging	Single-file DEX assumptions and incomplete scanners	Whole-APK extraction over all DEX files and split artifacts	Evaluation denominators must state whether all executable artifacts were analyzed.
Partial library inclusion	Binary present/absent decisions and complete-profile matching	Similarity scores, per-class evidence, explicit ambiguity classes	A partial match should not automatically become a full component/version claim.
Nested or transitive dependencies	Flat library lists and one-library-one-package assumptions	Dependency graphs, source-assisted validation, relationship-aware SBOM models	A recovered component may be direct, transitive, bundled, shaded, or copied; the relationship must remain uncertain unless evidenced.
Native and dynamic code	DEX-only static features	Native file hashes, load-site evidence, dynamic analysis, source validation	Out-of-scope code must be represented as a boundary, not silently ignored.

The most consequential threats for exact-version claims are shrinking, partial inclusion, and method inlining. Identifier renaming is comparatively well handled by modern structural or opcode-based approaches, but version evidence often depends on small differences between adjacent releases. Optimizations that delete, merge, or inline exactly those differences can turn an apparently confident match into a version-band ambiguity.

3.5 Representative Android TPL Approaches

3.5.1 From Profile Matching to Structural Boundaries

LibScout established a reference-profile model for matching compiled Android applications against known library releases. Its three-layer profile structure and fuzzy method descriptors made identifier renaming less damaging, and its similarity scores provided a graded alternative to exact profile equality [13]. From the viewpoint of trustworthy SBOM recovery, the important lesson is dual: reference profiles can support version claims, but those claims inherit the assumptions of the profile structure and the reference corpus. When package layout or method boundaries are transformed, the confidence of the resulting claim must be reduced rather than silently preserved.

LibD moved away from pure package-layout dependence by recovering library instances through a homogeny graph over inclusion, inheritance, and call relations, combined with opcode-derived fingerprints [16]. This made large-scale library discovery more resilient to naming changes, but the output remained a clustering-oriented library identification result rather than an exact-version or SBOM result. For the taxonomy, LibD is therefore strong evidence for graph-based boundary recovery and weak evidence for version authority.

ORLIS introduced a two-stage obfuscation-resilient design. Its first stage uses inter-procedural call-graph-derived features and `sdhash` to remove unlikely library candidates. Its second stage applies `ssdeep` class signatures and an injective class mapping to verify whether a library is included in the application [14]. The design is valuable because it separates scalable filtering from detailed confirmation and can represent partial inclusion without relying on package names. Its limitations are equally important for the proposed framework: a class mapping is not automatically an exact-version claim, and computational cost affects whether a method can be used as routine SBOM evidence.

LibPecker and LibID further illustrate the shift from names to structural and class-level evidence. LibPecker emphasizes class dependency and adaptive similarity. LibID formulates consistent class correspondence with binary integer programming and identifies specific library versions under code shrinking and package modification [17, 15]. Together with the later assessment studies, they support the review's central classification: package structure is a fragile feature family, while class relationships, signatures, and opcode-derived evidence are more suitable as identity support under obfuscation.

3.5.2 Version-Aware and Vulnerability-Oriented Pipelines

ATVHUNTER combines class-dependency based decoupling, coarse CFG filtering, fine-grained opcode matching, and a vulnerability database to identify vulnerable library versions [18]. It is one of the most important verified studies for this thesis because it connects version identification to security interpretation. However, its output remains a

version and vulnerability list rather than a standard SBOM with explicit provenance, uncertainty, and claim boundaries. The study therefore motivates the framework's rule that vulnerability authority cannot exceed the strength of the underlying version evidence.

LibScan and LibHunter extend the version-aware family by using richer opcode, class-correspondence, call-chain, and optimization-aware evidence [19, 20]. Their relevance is not only that they improve matching. They also show that code optimization itself is an evaluation dimension. In particular, LibHunter augments reference features to model call-site optimization and performs cross-inlining method matching. A method evaluated only on clear or lightly obfuscated applications has not demonstrated the same authority as a method evaluated under inlining and shrinking.

SAD provides the most detailed Tier A example of class-structural version-level reasoning. It splits classes into stateful and stateless categories, applies field-operation sequences or opcode similarity accordingly, and uses literals to break ties between adjacent versions [21]. This design supports the thesis claim that exact-version attribution requires evidence at a finer granularity than library presence. At the same time, its remaining sensitivity to method inlining confirms that strong reported version-level performance does not eliminate transformation risk.

3.5.3 Learning-Based Presence Detection

LibAttention represents a different family: instead of comparing explicit profiles, it learns representations from large Smali corpora and predicts library presence with a Transformer-based model [22]. This is important because learned features can capture patterns that rule-based signatures miss under aggressive obfuscation. For SBOM reconstruction, however, the reviewed Tier A result is bounded: it predicts presence rather than exact version, does not expose an explicit class or library-boundary mapping, and does not provide standard SBOM output. In the taxonomy, learned presence detection is therefore a potentially useful candidate-generation signal but not sufficient version or vulnerability authority by itself.

3.5.4 Empirical Assessments and Cross-Tool Studies

The ASE 2020 assessment and the later systematic assessment are not merely background surveys; they define several trust requirements for any recovered SBOM pipeline [11, 12]. They show that tool results depend on module decoupling, multidex handling, AAR processing, obfuscation setting, version-identification capability, efficiency, and ease of reproduction. These studies are a warning against reporting a single accuracy value without denominators, exclusions, and task levels.

The Ramirez thesis provides a cross-platform empirical comparison of Android and iOS library-detection tools [23]. Its Android study reuses 75 APKs from the ORLIS dataset, 75 from the LIBLOOM dataset, and 50 manually selected Google Play applications. It is therefore useful for cross-tool execution and dataset-reuse lessons, but it is not a source-known or F-Droid validation study and it does not independently establish standard SBOM output.

TPL-Benchmark strengthens the benchmark perspective with a large, version-annotated Android dataset and a standardized reevaluation of ten detection methods [24]. Its compatible tool-evaluation slice shows that the best library-level F1 score was 60.15%,

while LibHunter obtained the highest reported version-level recall at 76.48%. These results are not a universal ceiling, but they demonstrate that realistic shared ground truth can produce substantially weaker and more heterogeneous outcomes than isolated original evaluations.

UniBOM is an adjacent rather than Android-specific study. It combines firmware extraction, filesystem scanning, source-code analysis, SBOM generation, and vulnerability visualization, and evaluates the workflow on 258 wireless gateway firmware images and four IoT operating-system codebases [25]. It confirms that binary, metadata, and source evidence can coexist in one SBOM workflow. It does not, however, solve APK-side Maven exact-version attribution or provide ground truth for Android TPL detection.

3.6 Evaluation Practices and Ground Truth

The reviewed studies evaluate different questions even when they use similar words such as detection, identification, precision, recall, or version accuracy. A trustworthy synthesis must therefore classify evaluation practice before interpreting the metric.

Table 3.4: Evaluation-practice taxonomy in the Android TPL literature.

Evaluation practice	**What it can show**	**Boundary for SBOM trust**
Controlled library or reference corpus	Whether a matcher can distinguish known library profiles or versions under specified transformations	Results depend on reference completeness and may overstate performance on diverse real applications.
Synthetic or tool-based obfuscation tests	Whether selected transformations damage a feature family	Only covers the transformations and configurations tested; R8 inlining and shrinking require explicit treatment.
Market-scale discovery	Whether a detector scales and finds plausible library families in many apps	Without source ground truth, discovery volume is not the same as correctness or exact-version authority.
Open-source or source-known validation	Whether recovered outputs match dependency evidence available outside the APK	Strong for audited cases, but it is a validation lane, not proof that the same APK-only method is generally correct.
Cross-tool benchmark or assessment	Which tools fail under shared criteria and where methodology is inconsistent	Useful for gap analysis, but metrics remain bounded by the benchmark's dataset and task definitions.
Security or vulnerability mapping	Whether detected versions can be connected to vulnerability data	The vulnerability conclusion is valid only as far as the component identity, exact version, and vulnerability database mapping are valid.

This taxonomy explains why the chapter does not pool reported performance numbers. LibScout's profile precision, ORLIS's obfuscation precision, LibD's market-scale discovery volume, ATVHUNTER's version-level results, LibAttention's R8-resilient presence

detection, SAD's class-structural version scores, and TPL-Benchmark's cross-tool results answer different questions [13, 16, 14, 18, 22, 21, 24]. For a reconstructed SBOM, the evaluation question must be restated as: *which claim type is being validated, against which denominator, and under which transformations?*

3.7 Comparative Analysis

Table 3.5 applies the chapter taxonomy to the Tier A seed corpus. The final column states the trust interpretation rather than a simple score. This is deliberate: the same study can be strong evidence for one claim type and weak evidence for another. The table records only supported roles and limitations; missing study fields are not filled by inference.

Table 3.5: Tier A studies classified by recovery role and trust implication.

Study	Primary role	Main evidence and technique	SBOM-trust implication
LibScout [13]	Reference-profile identity and version matching	Fuzzy method descriptors, package-class-method profile hierarchy, similarity score	Supports the idea of version profiles, but package and method-structure assumptions must be exposed as evidence limits.
LibD [16]	Scalable library discovery and boundary recovery	Homogeny graph, call/inheritance/inclusion relations, opcode-derived fingerprints	Strengthens identity evidence under name obfuscation, but does not provide exact-version or SBOM authority.
LibPecker [17]	Class-dependency based detection	Class dependency hashing, adaptive similarity, weighted class matching	Shows that structural class relations are useful, while similarity still needs explicit thresholds and uncertainty handling.
ORLIS [14]	Obfuscation-resilient structural matching	Call-graph-derived coarse features with `sdhash`, followed by `ssdeep` class signatures and injective class mapping	Provides partial-inclusion evidence, but high cost and missing exact-version authority limit direct SBOM use.
LibID [15]	Obfuscated library and version identification	Consistent class correspondence through binary integer programming, with source-backed threshold evaluation	Supports specific-version claims under its tested transformations; downstream security authority still depends on reference and vulnerability evidence.

Study	Primary role	Main evidence and technique	SBOM-trust implication
ASE 2020 assessment [11]	Cross-tool failure analysis	Unified criteria, tool patches, multidex and AAR observations	Defines required evaluation dimensions and warns that engineering blind spots can dominate measured accuracy.
Systematic assessment [12]	Broader comparative methodology	Six evaluation criteria, module-decoupling comparison, tool availability and limitations	Supports taxonomy-based synthesis and shows that CDG-style evidence is preferable to package-only evidence.
ATVHUNTER [18]	Version and vulnerability-oriented detection	CDG decoupling, CFG coarse filter, opcode fuzzy matching, vulnerability DB	Strongly connects version evidence to security use, but still lacks standard SBOM provenance and full R8-inlining coverage.
LibScan [19]	Precise TPL identification	Class correspondence, method opcode similarity, call-chain evidence	Improves fine-grained matching evidence; claim strength remains tied to reference coverage and transformation conditions.
Ramirez thesis [23]	Cross-platform tool comparison	ORLIS and LIBLOOM Android datasets, manually selected Google Play apps, and Android/iOS contrast	Supports cross-tool and dataset-reuse analysis, but does not provide source-known Android ground truth or standard SBOM validation.
LibHunter [20]	Optimization-aware detection	Call-site-optimization feature enhancement and cross-inlining method matching	Makes optimization a first-class validity condition for exact-version claims.
LibAttention [22]	Learning-based presence detection	Transformer over Smali tokens and SDK structural vocabulary	Promising for obfuscation-resilient identity hints, but insufficient alone for exact versions, provenance, or CVE authority.
SAD [21]	Class-structural version-level detection	CDG neighbor aggregation, stateful/stateless split, field-operation sequence, literal tie-breakers	Provides strong version-discriminating evidence, while remaining sensitive to method inlining and lacking standard SBOM output.

Study	Primary role	Main evidence and technique	SBOM-trust implication
TPL-Benchmark [24]	Ground-truth benchmark and cross-tool reassessment	Version-annotated real applications, dependency extraction, and standardized evaluation of ten detectors	Provides realistic comparative evidence and exposes low or uneven generalization; it diagnoses detector limits rather than supplying a detector.
UniBOM [25]	Adjacent SBOM generation and analysis workflow	Firmware extraction, filesystem and source scanning, SBOM generation, CVE analysis, and visualization	Shows end-to-end SBOM integration outside Android TPL attribution, but cannot justify APK Maven identity or exact-version claims.

Three patterns emerge from this comparison. First, the strongest exact-version approaches are multi-evidence approaches: they do not rely on names alone but combine structural, opcode, class, and literal information. Second, evaluation authority is not the same as detection authority. Assessment studies and source-known validation do not identify libraries by themselves, but they determine whether a result can be interpreted safely. Third, standard SBOM trust requirements are largely external to the detector output: provenance, ambiguity, validation class, relationship type, and vulnerability authority are rarely represented as first-class fields.

3.8 Limitations of Existing Research

The taxonomy exposes limitations that recur across the verified literature:

1. **Library identity and exact version are often conflated.** Presence detection, class-set matching, and exact-version attribution are different claims, but tool outputs do not always preserve the distinction.

2. **Reference coverage is rarely treated as part of the result.** A missed match can result from absent references, extraction failure, or a weak matcher. Without coverage metadata, the user cannot tell which failure occurred.

3. **Transformation coverage is incomplete.** Identifier renaming is widely considered, but shrinking, partial inclusion, package flattening, and method inlining remain unevenly evaluated [20, 22, 21].

4. **Confidence is usually a threshold outcome, not an auditable explanation.** Many methods compute similarities or classifier scores, but few expose why a claim should be trusted, which signals conflicted, or why the method did not abstain.

5. **Partial libraries and dependency relationships are not fully represented.** Similarity scores can detect partial inclusion, but a standard SBOM also needs relationship semantics: direct, transitive, bundled, shaded, copied, or uncertain.

6. **Vulnerability interpretation is stronger than the underlying evidence in many pipelines.** A detected library name is not enough for a version-specific CVE claim. Even a plausible version candidate requires reference, transformation, and validation context.
7. **Reproducibility remains uneven.** Tool availability, installation difficulty, dataset access, and benchmark construction vary across studies, limiting independent confirmation of reported results.
8. **Standard SBOM output is not the dominant research artifact.** The verified studies mostly produce detector outputs rather than complete CycloneDX or SPDX statements with evidence, provenance, and uncertainty fields.

These limitations do not make the existing work weak. Rather, they show that the literature has mostly optimized detection performance, while trustworthy SBOM recovery requires an additional layer that explains what each recovered claim means and how it may be used.

3.9 Research Gap

The state of the art provides valuable techniques for recognizing Android third-party libraries from compiled artifacts. Reference profiles, fuzzy descriptors, dependency graphs, opcode features, call-chain evidence, stateful/stateless class summaries, learned Smali representations, and source-known validation all contribute useful evidence. Within the reviewed Android TPL literature and adjacent binary-SBOM studies, however, no verified study provides a unified model that connects these signals to a reconstructed SBOM claim while preserving all of the trust dimensions used in this thesis.

The research gap can therefore be stated as follows:

> Existing Android component detectors provide useful identification and version-discrimination techniques, but they do not consistently attach each recovered SBOM claim to its evidence provenance, reference coverage, transformation risk, confidence boundary, ambiguity, validation class, and permitted vulnerability interpretation.

This gap is not only a formatting problem. A syntactically valid SBOM can still be untrustworthy if it hides uncertainty, promotes identity evidence into exact-version authority, or turns an ambiguous version set into a definitive CVE finding. Conversely, a detector that abstains or reports a bounded version set may be more useful for security decision-making than a pipeline that always emits a single version without explaining the evidence.

3.10 Conclusion

This chapter classified Android component-recovery research by technique, evidence source, transformation threat, and evaluation practice. The synthesis shows a clear

technical progression from package and profile matching toward graph-based, version-aware, optimization-aware, and learning-based methods. It also shows a persistent trust gap: detector outputs rarely preserve enough provenance, uncertainty, validation context, and security authority to become auditable reconstructed SBOM claims on their own.

The next chapter responds to this gap by proposing an evidence-based trust framework. The framework does not replace the reviewed techniques. Instead, it defines how their outputs should be represented, fused, bounded, and interpreted so that a recovered component entry communicates not only what was detected, but why it should be believed and how safely it may be used.

Part IV

Proposed Framework

Chapter 4

Proposed Evidence-Based Trust Framework

4.1 Introduction

The previous chapter showed that Android third-party library detection has advanced from package-name heuristics toward structural, graph-based, optimization-aware, and learning-assisted methods. These techniques are valuable, but they do not by themselves answer the trust question required by an SBOM: when a compiled APK yields a component name, a candidate version, or a vulnerability warning, what exactly has been proven, what remains uncertain, and how safely may the result be used?

This chapter proposes an evidence-based trust framework for reconstructed SBOM claims. The framework is not a replacement for existing detection methods such as dependency-graph matching, opcode similarity, learned Smali features, or source-known validation [12, 18, 22, 21]. Instead, it defines a layer above those methods. That layer converts detector outputs into explicit claims with evidence, provenance, confidence boundaries, ambiguity, and permitted downstream interpretation.

The central idea is simple: every recovered component entry should be treated as a claim, not as an unqualified fact. A claim may be exact, bounded, identity-only, report-only, or out-of-scope. It may be supported by direct bytecode evidence, by independent structural evidence, by source validation, or only by contextual hints. It may also be limited by missing references, partial inclusion, R8 optimization, dependency shading, native code, or hidden runtime loading. The framework makes these distinctions visible so that an SBOM user can understand not only what was emitted, but why it was emitted and what it is allowed to mean.

4.2 Framework Design Principles

The framework is built around ten design principles. They are intentionally conservative because reconstructed SBOMs are often used for security decisions, compliance checks, and vulnerability triage.

P1. Claim explicitness. Every emitted component, version, relationship, or vulnerability finding is represented as a claim with an identifier, a claim

type, supporting evidence, and a current state. The pipeline does not silently transform a similarity score into a fact.

P2. Identity-version separation. Component identity and exact version are separate claims. Evidence that supports the presence of `okhttp` or `androidx.core` does not automatically support a specific version. This separation is essential for CVE interpretation because most advisories are version-dependent.

P3. Evidence provenance. Each signal records where it came from: APK-side bytecode, manifest metadata, resource evidence, reference database profiles, source-assisted validation, external package metadata, vulnerability databases, or report-only model output.

P4. Reference coverage awareness. A negative or weak result is interpreted in relation to the available reference database. Missing coordinates, missing versions, failed feature extraction, and incomplete profile families are treated as coverage risks, not as proof that a component is absent.

P5. Transformation-bounded confidence. Confidence is capped when the analyzed artifact is affected by transformations that the evidence cannot reliably handle, including package renaming, code shrinking, partial inclusion, repackaging, multidex splitting, class merging, and method inlining [9, 20].

P6. Conflict preservation. Competing versions, adjacent library families, or disagreeing evidence sources remain visible. The framework prefers a bounded ambiguity set or an abstention over a false sense of exactness.

P7. Validation-lane separation. Controlled fixtures, clean source-validated real applications, partial source-validated applications, source-known upper bounds, diagnostic experiments, and limitation cases are separate lanes. Scores from one lane must not be used as if they prove another lane.

P8. Authority monotonicity. Downstream authority cannot exceed upstream evidence. A vulnerability claim cannot be stronger than the component and version claim on which it depends; a standard CycloneDX or SPDX entry cannot hide unresolved ambiguity behind a syntactically valid package URL [4, 5, 6].

P9. Report-only containment. Diagnostic or machine-learning assistance may guide investigation, ranking, and future work, but it is not allowed to modify authoritative component, version, or vulnerability outputs until it passes the same validation and regression gates as the deterministic evidence.

P10. Auditable abstention. Not emitting a component, emitting identity without version, or emitting a version band is a legitimate outcome when evidence is insufficient. The framework records why the pipeline abstained so that uncertainty becomes a visible result rather than a silent failure.

4.3 Recovered Claim Model

4.3.1 Claim Definition

A recovered claim is a structured statement about a component-related fact in an analyzed APK. The claim may concern component identity, exact version, version range, artifact family, dependency relationship, file evidence, or vulnerability exposure. Formally, a claim can be described as:

$$C = \langle s, t, a, e, p, r, q, u, v, \alpha \rangle$$

where s is the analyzed subject, t is the claim type, a is the asserted value, e is the supporting evidence set, p is provenance, r is reference and transformation risk, q is the evidence quality profile, u is uncertainty or ambiguity, v is validation context, and α is the permitted downstream authority. This notation is not meant to impose a specific storage format. It is a discipline for preventing weak evidence from being consumed as a stronger result.

Table 4.1: Core fields of a reconstructed SBOM claim.

Field	Question answered	Examples
Subject	What artifact is being analyzed?	APK identifier, package name, hash, build variant, DEX or native file scope.
Claim type	What kind of statement is this?	Component identity, exact version, bounded version set, relationship, file component, vulnerability status.
Asserted value	What is being claimed?	Maven coordinate, PURL, version, candidate family, dependency edge, native file hash.
Evidence set	Why is the claim supported?	Class dependency graph, opcode/literal features, package metadata, reference profile match, source validation, advisory match.
Provenance	Where did the evidence come from?	APK-side extraction, reference database, source repository validation, external vulnerability source, report-only model.
Coverage context	What references were available?	Present or missing coordinate versions, artifact-family coverage, extraction failures, source resolution status.
Transformation risk	What may have damaged the signal?	R8 shrinking, method inlining, package flattening, partial inclusion, multidex, hidden DEX, native payloads.
Quality profile	How reliable is the evidence?	Directness, uniqueness, independence, reproducibility, version specificity.
Uncertainty	What is unresolved?	Ambiguous adjacent versions, family-level match only, missing-source component, unsupported native package version.
Validation context	Where has the claim been checked?	Controlled fixture, clean real APK, partial real APK, source-known upper bound, diagnostic lane.
Permitted authority	How may it be used?	Confirmed SBOM component, potential component, report-only hint, confirmed vulnerability, potential vulnerability, out-of-scope boundary.

4.3.2 Claim Granularity

The framework distinguishes four levels of component granularity:

1. **Identity claim:** a component family or coordinate is present, but the exact version is not established.
2. **Version claim:** a specific version or bounded version set is supported by version-discriminating evidence.
3. **Relationship claim:** the component is direct, transitive, bundled, copied, shaded, partially included, or uncertain.
4. **Security claim:** a vulnerability advisory applies, may apply, or does not apply under the current evidence.

This granularity is important because SBOM formats can represent package entries even when the recovery process is uncertain. The framework therefore separates syntactic representation from epistemic authority. A package URL can identify a candidate component, but the confidence class explains whether the PURL is exact, bounded, or only investigatory.

4.4 Evidence Hierarchy

The evidence hierarchy ranks evidence by what it can safely support. It is not a universal probability scale. Instead, it is a policy for deciding which interpretations are permitted when the available evidence is incomplete or conflicted.

Table 4.2: Evidence hierarchy for reconstructed component claims.

Level	Meaning	Typical evidence	Permitted interpretation
E0	No usable component evidence	No matching code, missing extraction, unsupported payload type, or unavailable reference family	No package-level claim. The result may record a boundary or extraction failure.
E1	Weak contextual evidence	Package-name fragment, manifest hint, resource string, file name, or nearby ecosystem cue	Investigation hint only; not enough for an SBOM component claim.
E2	Identity-supporting evidence	Class-set overlap, namespace pattern, partial structural match, or learned presence signal	Candidate component identity; exact version and vulnerability authority remain unavailable.
E3	Independent multi-source identity evidence	Agreement between structural graph evidence, opcode/literal evidence, reference family metadata, or independent detector families	Strong identity claim, but exact version still requires version-discriminating evidence.
E4	Version-discriminating APK evidence	Version-specific graph, opcode, literal, API, or stateful/stateless class features that separate nearby versions	Exact version or bounded version set, subject to conflict, coverage, and transformation caps.
E5	Externally validated exact evidence	Source-resolved dependency tree, controlled fixture ground truth, reproducible build evidence, or manually verified exact-version evidence	Confirmed exact claim within the validation context; still not proof of global detector completeness.

Two aspects of the hierarchy are deliberately strict. First, evidence levels are claim-specific. A signal may be E3 for identity but only E1 for exact version. Second, external

validation does not erase the conditions under which it was obtained. A controlled fixture can validate pipeline mechanics, while a source-validated real application can validate emitted versions for that application. Neither automatically proves that all real-world dependencies are recoverable.

4.5 Evidence Quality Dimensions

Evidence level describes what a signal may support. Evidence quality describes how fragile that support is. The framework evaluates each evidence item along the dimensions in table 4.3.

Table 4.3: Evidence quality dimensions used to bound claim confidence.

Dimension	Trust question	Examples of risk
Directness	Does the signal observe the component itself or only context around it?	Manifest strings and package names are indirect; class and method structure are more direct.
Uniqueness	Does the signal distinguish one component or version from close neighbors?	Shared utility classes, shaded code, and adjacent AndroidX versions reduce uniqueness.
Independence	Do multiple signals fail together or independently?	Two features derived from the same package namespace are weaker than namespace plus graph plus literal evidence.
Version specificity	Can the evidence separate exact versions?	Identity features may confirm a coordinate while still leaving a version band.
Transformation resilience	Does the signal survive obfuscation and optimization?	R8 shrinking, inlining, class merging, and package flattening can remove or distort features.
Reference coverage	Were the relevant reference versions available and successfully extracted?	Missing historical versions or failed profiles cap confidence even when the best available candidate scores highly.
Provenance reliability	Is the evidence reproducible and auditable?	Manually curated notes, source-resolution logs, and generated feature artifacts carry different audit strength.
Conflict level	Do plausible alternatives remain?	Near-ties, artifact family ambiguity, and disagreement between source and APK-side evidence require explicit conflict handling.

These dimensions allow a detector to be useful without being overconfident. For example, a learned Smali classifier may provide transformation-resilient identity evidence, but if it lacks exact-version specificity and independent validation, the framework keeps it report-only or identity-only. Conversely, source validation can strongly confirm emitted versions for one real application, but if the source tree contains dependencies that do not

appear in the compiled artifact, the result remains an emitted-version confirmation rate rather than full dependency recall.

4.6 Evidence Fusion, Conflict, and Abstention

4.6.1 Fusion Policy

The framework treats evidence fusion as an auditable decision process rather than a single opaque score. A practical implementation may still use numeric similarity scores, classifier outputs, and thresholds, but the final claim is assigned by policy. Formal evidence theories provide richer mathematical fusion rules [26]; the present framework does not claim to implement Dempster–Shafer inference and instead uses an explicit, provenance-preserving authority policy:

$$\text{trust}(C) = f(S, D, I, V) - g(K, R, T)$$

This expression is conceptual. It should not be read as a calibrated probability unless a separate calibration study has been performed. Its role is to make the monotonicity rule explicit: more independent, direct, and version-specific evidence can increase trust, while unresolved conflict, missing references, and transformation risk can only cap or reduce it. This caution is necessary because even high-performing learned scores can be poorly calibrated [27]. Here, S, D, I, and V denote support, directness, independence, and version specificity; K, R, and T denote conflict, reference coverage risk, and transformation risk.

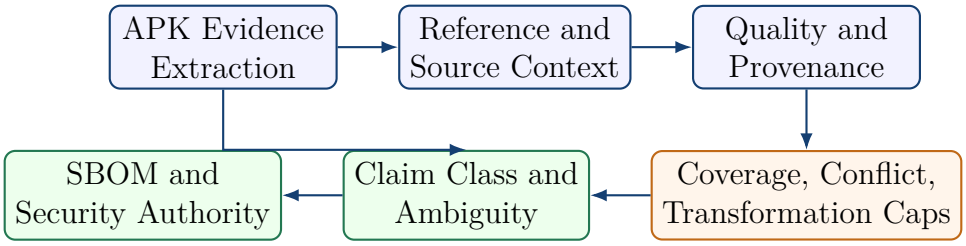


Figure 4.1: Conceptual flow of the proposed evidence-based trust framework.

4.6.2 Conflict Handling

Conflict is not an exceptional case in APK-side SBOM recovery; it is normal. Adjacent versions may share most bytecode, libraries may be partially bundled, old source trees may resolve different dependency versions from current reference repositories, and optimized builds may remove the features that would distinguish two candidates. The framework therefore defines the following conflict actions:

1. **Exact promotion:** promote a single exact version only when the selected candidate has version-discriminating evidence and plausible alternatives are ruled out by independent signals or validation.
2. **Bounded ambiguity:** emit a version set or band when the evidence supports a component family but cannot safely separate nearby versions.

3. **Identity-only output:** emit the component identity without a version when version evidence is weak, missing, or transformation-damaged.
4. **Report-only output:** preserve diagnostic candidates for human review or future experiments without changing authoritative SBOM fields.
5. **Abstention:** record no component claim when the evidence does not meet the minimum identity threshold or when the payload is outside the method's scope.

4.6.3 Confidence Caps

Confidence capping is the key mechanism that prevents overclaiming. A claim cannot exceed the weakest relevant boundary. For example, if the best available reference database lacks the source version, an exact-version match against a newer nearby reference may remain a candidate rather than a confirmed result. If a detector has only report-only validation, its result cannot modify authoritative CycloneDX components. If a native shared object is observed but no native package-version resolver exists, the framework may emit a file component with hash evidence but not a Maven or native package version claim.

4.7 Claim Classes

Table 4.4 defines the claim classes used by the framework. They are ordered by permitted authority, not by how useful they are. An identity-only or bounded claim can be more honest, and therefore more useful, than a false exact claim.

Table 4.4: Claim classes and permitted downstream use.

Claim class	Required support	Permitted use
Confirmed exact	E5 validation or E4 evidence with strong independent corroboration and no unresolved nearby conflict	Authoritative SBOM component and exact-version-dependent security interpretation.
Strong exact	E4 version-discriminating evidence with adequate reference coverage and manageable transformation risk	Authoritative component entry with exact version, but validation context must be shown.
Likely exact	E4 evidence exists, but one quality dimension is weak or a limited alternative remains	Component entry may be emitted with confidence caveat; vulnerability findings should be conservative.
Version-band ambiguous	Identity is strong, but exact version cannot be separated from nearby candidates	Emit component with bounded version set or ambiguity field; exact CVE classification is not allowed without additional policy.
Identity-only	E2 or E3 identity evidence without version-discriminating support	Emit candidate component identity; do not emit exact-version CVE conclusions.
Report-only candidate	Diagnostic, experimental, or machine-learning evidence that has not passed authority gates	Human investigation, prioritization, or future work only.
Unsupported or abstained	Evidence below identity threshold or conflict too severe	No package claim; record reason for abstention if useful.
Boundary or out-of-scope	Payload is observed, but the method does not support its package or version semantics	File/hash component, limitation note, or risk flag rather than package-version recovery.

4.8 Validation and Evidence Lanes

Evaluation results are meaningful only when their lane is explicit. The framework separates the lanes in table 4.5 so that strong evidence in one context is not accidentally generalized to another.

Table 4.5: Validation and evidence lanes.

Lane	What it can show	What it must not imply
Controlled ground truth	Pipeline mechanics, exact expected versions, known transformations, and regression behavior under designed fixtures	Not a measure of real-world dependency completeness.

Lane	What it can show	What it must not imply
Clean source-validated real APKs	Exact emitted Maven versions can be checked against source-resolved dependency evidence for real applications	Not a global real-world score if the clean set is small or sparse.
Partial source-validated real APKs	Harder real applications can be analyzed with exact matches, mismatches, missing-source rows, and ambiguity separated	Partial evidence is not a total failure, but it is also not full recall.
Source-known upper bound	Source trees can indicate dependencies that may or may not be compiled into the APK	Source dependency presence alone does not prove APK-side presence.
Report-only diagnostic lane	Experimental signals can be inspected without changing authoritative results	Diagnostic improvement is not permission to alter SBOM or CVE outputs.
Experiment-only optimization lane	Narrow fixes can be tested with controls, near-tie analysis, and regression checks	A scoped preflight is not a shared default until controls and evidence gates pass.
Limitation or boundary lane	Native payloads, hidden code, unsupported transformations, and extraction failures can be reported honestly	Boundary cases must not be counted as Maven-version failures or successes.

This lane structure also changes how metrics should be reported. A controlled fixture success rate, a clean real-APK emitted-version confirmation rate, and a partial real-APK confirmation rate answer different questions. Combining them into one number can be useful only after the meaning and denominator are made explicit. In particular, an emitted Maven version confirmation rate should not be described as full dependency recall unless the evaluation has shown that all relevant source dependencies are present in the compiled artifact and within scope.

4.9 Vulnerability-Claim Policy

CVE and advisory mapping is downstream of component recovery. The framework therefore applies an authority rule: vulnerability status cannot be stronger than the recovered component and version evidence. This is especially important when advisory ranges are narrow, when adjacent versions have different security status, or when the result is identity-only. The policy is compatible with VEX-style separation between component inventory and the status or applicability of a vulnerability claim [28].

Table 4.6: Vulnerability interpretation policy.

Component evidence	Allowed vulnerability status	Rationale
Confirmed exact vulnerable version	Confirmed affected	The version claim has sufficient authority to evaluate version-specific advisory ranges.
Confirmed exact non-affected version	Confirmed not affected for that advisory	The result is advisory-specific; absence of one advisory does not prove the component is risk-free.
Likely exact version	Potential affected or likely affected, depending on policy	The version may be used for triage, but the confidence caveat remains visible.
Version-band ambiguous	Potential affected if any candidate in the band is affected	Exact affected/not-affected labels are unsafe unless all candidates share the same advisory status.
Identity-only component	Potential exposure only when advisory identity matches and version is unknown	Version-specific claims are not permitted.
Report-only candidate	Report-only vulnerability hint	The finding may guide investigation but must not change authoritative security status.
Boundary or out-of-scope payload	File/hash risk note or external analysis required	Package-version advisory mapping is unavailable in the current method.

This policy supports conservative security reporting. It avoids two common errors: hiding a possible issue because the exact version is unknown, and declaring a confirmed vulnerability from evidence that only supports component identity. The result can still be actionable: a potential finding may trigger manual review, dependency upgrade investigation, or source validation, while a confirmed finding can support stronger remediation language.

4.10 Recovered Claim Lifecycle

The framework can be implemented as a claim lifecycle with explicit state transitions:

1. **Extract evidence** from the APK, including bytecode, resources, manifests, native files, and available build metadata.
2. **Generate candidates** from reference profiles, structural features, learned hints, package metadata, or source-known context.
3. **Normalize identifiers** into coordinates, package URLs, file hashes, advisory identifiers, and relationship candidates.
4. **Attach provenance** for every evidence item and candidate source.

5. **Assess quality and coverage** by checking directness, independence, version specificity, reference availability, and transformation exposure.
6. **Fuse evidence** while preserving conflicts, near-ties, and missing-reference explanations.
7. **Assign claim class** with confidence, ambiguity, validation lane, and permitted downstream authority.
8. **Emit or abstain** in the SBOM, vulnerability report, and diagnostic artifacts according to the assigned authority.

The lifecycle is intentionally auditable. A reviewer should be able to trace a CycloneDX component entry or vulnerability row back to the features, reference profiles, source validation, and policy decisions that produced it.

4.11 PFE System as a Case Study

The PFE implementation associated with this thesis provides a concrete case study for the framework. The goal of this section is not to present final implementation evaluation as a second master experiment. Instead, the mapping shows how the proposed trust model organizes evidence from a real APK-to-SBOM pipeline, and the following section evaluates whether the model preserves the project's established evidence lanes and authority boundaries.

Table 4.7: Mapping PFE observations to framework concepts.

Case-study observation	Framework concept	Safe interpretation
Controlled transformed fixtures with known expected Maven versions	Controlled ground-truth lane and E5 validation	Demonstrates pipeline mechanics and transformation-specific regressions under designed conditions; does not prove complete real-world recovery.
Clean source-validated real APKs with exact emitted Maven-version checks	Clean real-APK validation lane	Confirms emitted versions for those applications and denominators; should remain separate from partial and sparse cases.
Partial source-validated real APKs with exact matches, version mismatches, and missing-source rows separated	Partial real-APK lane, conflict preservation, and emitted-version confirmation rate	Shows useful recovery in difficult applications without presenting unresolved rows as either success or total failure.

Case-study observation	**Framework concept**	**Safe interpretation**
Reference coverage and matcher recall measured as separate gaps	Reference availability, candidate-generation coverage, and denominator separation	Shows whether failure arises because the true release is absent from the reference base or because available evidence was not selected; neither gap should be hidden inside one accuracy value.
Source-known constrained reruns with exact source versions available	Source-known upper-bound lane	Demonstrates recognition under a favorable candidate space; does not measure unconstrained APK-only accuracy or prove that every source dependency survives in the binary.
Version-band ambiguity and adjacent-version near-ties	Bounded ambiguity, confidence caps, and experiment-only optimization lane	Prevents a weak near-tie from becoming a confirmed exact version or a confirmed CVE status.
Broad scoring and reference-database changes that improved targets but regressed controls	Negative evidence and promotion governance	Preserves failed experiments as evidence that a signal is unsafe for shared authority, even when it is useful diagnostically.
The scoped AndroidX Fragment 1.6.2 preflight selected the intended release without sampled control regressions, but lacked version-unique APK evidence	Experiment-only near-tie decision	Keeps the result outside the shared reference database because clean sampled controls do not convert feature-equivalent candidates into exact evidence.
Report-only R8 and AI assistance	Report-only containment	Allows learned or diagnostic signals to support triage while keeping authoritative SBOM and vulnerability outputs deterministic until validation gates pass.
Native shared objects and hidden-code cases	Boundary or out-of-scope claim class	Supports file/hash evidence, limitation notes, or risk flags without inventing Maven or native package-version claims.

This mapping illustrates the main contribution of the framework: it turns mixed evidence into bounded statements. A controlled result can be strong without being overgeneralized. A partial real-APK result can be valuable without being hidden as a failure. A source-known dependency can inform an upper bound without proving APK-side presence. A report-only classifier can guide investigation without becoming SBOM authority. These distinctions make the pipeline more cautious, but they also make it more usable because users can decide how to act on each claim class.

4.12 Framework Case-Study Evaluation

The framework was evaluated against the audited PFE evidence package at two levels: representational coverage and authority preservation. Representational coverage asks whether every major project outcome can be expressed without collapsing its validation context. Authority preservation asks whether weak, conflicting, source-constrained, or learned evidence remains prevented from silently changing authoritative component, version, or vulnerability claims. The underlying ledgers are retained in the project artifacts, including `docs/evaluation/THESIS_EVALUATION_TABLES.md`, `docs/thesis/FINAL_V2_RESULTS_SECTION.md`, and the experiment-specific diagnostic reports.

Table 4.8: Case-study evaluation of representational coverage and authority preservation.

Criterion	Observed PFE evidence	Framework evaluation
Evidence-lane separation	The controlled lane confirmed 84/84 expected versions. The clean real lane confirmed 47/47 emitted versions, while the partial real lane confirmed 880/1790. Four explicit boundary cases were excluded from accuracy metrics.	Passed for the case study: strong, partial, and boundary evidence remained separate rather than being pooled into one success claim.
Denominator separation	Across 102 source-validated real APKs, 927/1837 emitted Maven versions were confirmed. Against the wider 12,082-row source dependency denominator, the same 927 exact emitted rows represent 7.7% end-to-end exact source recall.	Passed: emitted-row correctness and full source-dependency recall are stated as different metrics with different authority.
Upper-bound containment	In 106 source-known constrained cases, 6772/6772 emitted rows matched the source versions, while 5238 source rows remained gaps.	Passed: the result is represented as a favorable source-known recognition upper bound, not as unconstrained APK-only accuracy.
Conflict and negative controls	The Fragment 1.6.2 preflight selected the intended version in 10/10 target rows but remained an unresolved near-tie. The coroutines 1.6.4 experiment fixed 20/20 target rows but regressed three controls. Neither change was promoted.	Passed: apparent target improvements did not override ambiguity or negative control evidence.
Report-only containment	The AI/R8 corpus contains 31,589 rows, with 20,523 training-eligible and 11,066 held-out or report-only rows. Report-only integration left normal outputs unchanged, and stricter APK-only gates produced zero promotion candidates.	Passed: learned assistance can support triage while component selection, SBOM emission, and vulnerability authority remain unchanged.

Criterion	Observed PFE evidence	Framework evaluation
Transformation-risk containment	The Phase 4 multi-scale R8 diagnostic found ten possible real improvements, but also six controlled regression blockers, 121 real regression-risk rows, and ten clean-real blockers. Global promotion was rejected.	Passed: transformation-oriented evidence remained diagnostic when its control matrix did not support shared authority.
Boundary representation	Four native-heavy or obfuscation cases remained outside accuracy metrics. Seventeen native files were represented as hash-only evidence; the hybrid Jitsi case validated 5/7 Maven rows while keeping 13 native files hash-only.	Passed: unsupported package/version claims were not invented for native, hidden, or framework-specific payloads.

The evaluation therefore supports a bounded conclusion: the framework covers all major evidence outcomes exercised by the PFE and preserved authority in the observed negative-promotion, report-only, and boundary cases. It does not establish calibrated confidence probabilities, completeness across all Android detectors, or independent analyst agreement. Those questions require external datasets, tools, and reviewers. The case study nevertheless replaces a purely proposed evaluation with auditable evidence that the framework can describe real successes, failures, near-ties, upper bounds, and abstentions without changing their meaning.

4.13 Limitations and Research Agenda

The proposed framework is intentionally evidence-centric, but it has several limitations.

1. **It does not solve feature extraction by itself.** The framework requires detectors, reference profiles, source validation, and vulnerability data. Weak upstream evidence still limits the final claims.

2. **It does not provide calibrated probabilities.** Confidence classes are policy categories unless supported by a separate calibration study over representative datasets.

3. **It depends on reference and source availability.** Missing Maven versions, unavailable historical artifacts, broken build systems, and incomplete source trees can cap evidence even when the APK-side matcher is strong.

4. **It does not fully solve transitive dependency semantics.** Reconstructed bytecode can show that code is present, but distinguishing direct, transitive, shaded, copied, and build-time-only dependencies remains difficult.

5. **It leaves native package-version recovery as future work.** Native files can be represented as file components with hashes, but mapping them to precise upstream package versions requires a different evidence base.

6. **It requires disciplined reporting.** The framework is only useful if evaluation

tables, thesis claims, and generated reports keep the lanes and authority boundaries visible.

These limitations define a research agenda. Future work should combine multi-scale bytecode evidence with calibrated confidence models, richer reference coverage, reproducible source-validation pipelines, native library package recovery, and benchmark protocols that measure not only exact matches but also abstention quality and vulnerability-authority correctness.

4.14 Conclusion

This chapter proposed an evidence-based trust framework for reconstructing SBOM claims from compiled Android applications. The framework treats each component, version, relationship, and vulnerability statement as an auditable claim with evidence, provenance, quality dimensions, validation context, ambiguity, and permitted downstream authority. Its main purpose is to prevent weak or partial evidence from being consumed as stronger security knowledge.

The framework also provides a bridge between the literature reviewed in the previous chapter and the PFE APK-to-SBOM implementation. Existing detection techniques supply useful signals; the framework defines how those signals are bounded, fused, and reported. This makes trustworthy SBOM recovery less a question of always producing one exact answer and more a question of producing the strongest answer that the evidence can honestly support.

Part V

General Conclusion

Chapter 5

General Conclusion and Future Directions

5.1 General Conclusion

Recovering an SBOM from a compiled Android application is not only a detection problem. It is a problem of deciding what can be claimed from incomplete, transformed, and sometimes conflicting evidence. A component name or exact version may look precise while depending on weak namespace overlap, an incomplete reference collection, a feature-equivalent adjacent release, or source information that does not prove binary presence. Trust therefore depends on how evidence is qualified and bounded, not merely on whether a tool returns an answer.

Chapter 2 established the distinction between build-generated and reconstructed SBOMs and defined the methodological conditions for reviewing binary component-recovery research. The taxonomy in Chapter 3 then organized the literature by recovery role, evidence source, transformation exposure, version capability, validation practice, and trust implication. This synthesis showed that the field offers many useful technical signals, but does not provide one evidence model that connects those signals to provenance, ambiguity, validation context, and permitted security interpretation.

Chapter 4 addressed that gap with an evidence-based trust framework. The framework treats every recovered component, version, relationship, and vulnerability statement as a structured claim. Each claim records its evidence, provenance, coverage, transformation exposure, confidence class, unresolved alternatives, validation lane, and downstream authority. Evidence may be fused when it is sufficiently independent and claim-relevant, but authority is capped by missing references, unresolved conflicts, weak version specificity, and unsupported payload types. Abstention is therefore a valid result rather than an implementation failure.

The associated PFE system was used as a bounded case study, not as a second set of master experiments. The evaluation confirmed 84/84 controlled versions, 47/47 emitted versions in the clean real-APK lane, and 880/1790 in the partial lane, while keeping four boundary cases outside accuracy metrics. It also separated the 927/1837 emitted-version confirmation result from the 927/12,082 full source-dependency denominator, contained a 6772/6772 source-known upper bound, rejected target improvements that failed ambiguity or control gates, and kept AI/R8 assistance report-only. These outcomes

support representational coverage and authority preservation within the case study while leaving broader independent validation as future work.

The central answer of this thesis is consequently conditional: heterogeneous APK evidence can support trustworthy reconstructed SBOM claims when the system preserves provenance, evaluates evidence at the identity and version levels separately, exposes conflict and missing coverage, distinguishes validation lanes, and prevents downstream vulnerability authority from exceeding the underlying component claim. Trustworthiness is not synonymous with always selecting an exact version. It is the ability to produce the strongest claim that the available evidence can justify, while making the remaining uncertainty visible and actionable.

5.2 Answers to the Research Questions

Table 5.1 summarizes the answers to the six supporting research questions introduced in Chapter 1.

Table 5.1: Answers to the supporting research questions.

RQ	Question focus	Answer
RQ1	Evidence sources available in compiled Android applications	Recoverable evidence spans archive and manifest metadata, DEX package and class structure, methods, opcodes, literals, API references, dependency and call relations, resources, native-file observations, versioned reference artifacts, publication metadata, and external validation such as reproducible fixtures or source-resolved dependency trees. These sources differ in directness, version specificity, resilience, and authority.
RQ2	How existing approaches combine evidence	Existing approaches range from namespace and class-set matching to structural graphs, opcode and semantic features, fuzzy signatures, learned representations, metadata coherence, and multi-stage reference matching. The literature supports multi-source reasoning, but no combination is universally reliable under shrinking, optimization, repackaging, feature-equivalent releases, and incomplete references. Fusion must therefore remain explicit, scoped, and subject to controls.
RQ3	Factors that determine identity and version strength	Evidence strength depends on directness, uniqueness, independence, transformation resilience, version specificity, provenance reliability, reference coverage, and reproducibility. Identity and exact-version support must be assessed separately because a signal can identify a library family while remaining unable to distinguish adjacent releases.

RQ	Question focus	Answer
RQ4	Representation of conflict, absence, transformation, and source dependence	The framework represents these conditions through competing candidates, bounded version sets, conflict records, coverage and transformation caps, explicit validation lanes, report-only outcomes, boundary classes, and abstention. Missing evidence is not interpreted as component absence, and source-known dependencies are not treated as proof of APK-side presence.
RQ5	Information required for an auditable and security-relevant claim	A claim must identify the subject APK, candidate component and artifact family, claim type and granularity, version or version set, supporting and opposing evidence, provenance, coverage, transformation exposure, confidence class, ambiguity, validation context, and permitted downstream use. Vulnerability interpretation must inherit these boundaries rather than bypass them.
RQ6	Evaluation without overstating accuracy	Evaluation should separate controlled, clean source-validated, partial source-validated, source-known upper-bound, diagnostic, experiment-only, and limitation lanes. It should publish the relevant denominator, distinguish emitted-row correctness from full dependency recall, retain negative experiments and abstentions, test traceability and authority preservation, and require target/control evidence before promoting a signal. In the PFE case study, these rules preserved all major evidence lanes and blocked unsafe promotions despite apparent target improvements.

Together, these answers shift evaluation from the question "Did the detector emit a version?" to the stronger question "What evidence supports this claim, what could invalidate it, and how may it safely be used?"

5.3 Main Contributions

The thesis makes six connected contributions.

1. **A recovery-oriented taxonomy.** The reviewed work is organized by recovery objective, evidence family, transformation threat, version capability, evaluation practice, and resulting trust implication rather than by headline performance alone.
2. **Operational trust requirements.** The thesis defines trustworthy reconstruction through provenance, evidence visibility, identity/version separation, uncertainty preservation, validation context, bounded vulnerability authority, and explicit denominators.
3. **A structured recovered-claim model.** Component identities, exact or bounded

versions, relationships, native observations, and vulnerability statements are represented as auditable claims with evidence and permitted authority.

4. **An evidence hierarchy and fusion policy.** Evidence is ranked by what it may support, evaluated across quality dimensions, fused only under independence and relevance constraints, and capped when conflicts, transformation risk, or missing coverage remain.
5. **A validation and vulnerability-authority model.** The framework separates evidence lanes and ensures that a vulnerability statement cannot be stronger than the component identity and version evidence from which it is derived.
6. **An evaluated PFE-informed framework case study.** Audited project observations show that the model represents exact success, partial recovery, source-known upper bounds, near-ties, failed promotions, report-only assistance, and unsupported native or hidden-code regions without presenting them as equivalent evidence or increasing their downstream authority.

The conceptual contribution is independent of the PFE architecture. A future detector could replace the current feature extractor, matcher, or learned assistant while retaining the same claim schema, evidence-quality questions, validation lanes, and authority rules.

5.4 Limitations

5.4.1 Review Scope and Evidence Availability

The structured review is concentrated on Android and closely related binary component-recovery research. The 15-study Tier A corpus is grounded in locally available full text, while five Tier B records remain useful discovery evidence but require additional verification before supporting strong comparative claims. The review should therefore be understood as structured and reproducible within its recorded scope, not as proof that every relevant publication has been captured.

5.4.2 Domain and Ecosystem Scope

The framework is derived mainly from DEX, Maven, Android build behavior, and the evidence problems encountered in compiled APKs. Its general principles may transfer to other ecosystems, but evidence hierarchies and validation rules would need adaptation for native packages, JavaScript bundles, containers, firmware, or platform-specific package managers.

5.4.3 Uncalibrated Confidence Classes

The proposed confidence and evidence levels are policy categories, not universally calibrated probabilities. They define permitted interpretations and useful abstention behavior, but do not state that a class has the same empirical error rate across component families, application eras, or transformation profiles.

5.4.4 Case-Study Dependence

The PFE provides varied and traceable scenarios, yet it was designed by the same project context that informed the framework. This creates a risk that the claim classes fit the available artifacts especially well. Independent tools, datasets, and analyst studies are needed to test whether the framework remains complete, understandable, and useful outside this implementation.

5.4.5 Source and Reference Boundaries

External source validation is strong evidence for declared or resolved dependencies, but it is not a perfect manifest of code present in the APK. Shrinking, optional dependencies, metadata-only artifacts, vendoring, and build-time tooling complicate the relation between source trees and binary content. Likewise, a reference-based detector cannot identify an exact release that is absent from its reference base.

5.4.6 Native, Dynamic, and Protected Code

The framework can represent native files, hidden payload indicators, and unsupported transformations as boundary evidence. It does not solve native package/version attribution, runtime-loaded dependency inventory, encrypted payload recovery, or commercial protector analysis. These areas require different evidence sources and validation protocols.

5.5 Future Research

Table 5.2: Priority directions for extending the trust framework.

Direction	Research objective	Required evidence
Empirical calibration	Calibrate identity, exact-version, ambiguity, and abstention classes by component family and transformation profile	Held-out ground truth, reliability analysis, and error-sensitive calibration metrics.
Independent framework validation	Apply the claim model and authority policy to other detectors, analysts, and datasets	External tools, blinded case review, and inter-rater agreement on claim classes.
Public benchmark design	Measure exactness, coverage, abstention, provenance, and vulnerability-authority correctness together	Reproducible APKs, complete denominators, transformation labels, and version-discriminating ground truth.
Provenance-aware SBOM profiles	Standardize how reconstructed claims expose evidence, ambiguity, and validation context	Interoperable CycloneDX or related profiles tested across producers and consumers.

Direction	Research objective	Required evidence
Evidence knowledge graphs	Connect coordinates, aliases, target variants, releases, binary evidence, source observations, and vulnerability ranges	Provenance-preserving graph schemas with controls against metadata-only overconfidence.
Native and dynamic attribution	Extend claims to shared libraries, bundled runtimes, protected payloads, and downloaded code	Separate corpora, architecture-aware fingerprints, runtime provenance, and scope-specific metrics.
Human factors and explainability	Determine whether analysts understand and act appropriately on claim classes, conflicts, and abstentions	User studies, decision-quality measures, and explanation comparisons.

5.5.1 Calibrated and Set-Valued Claims

A first priority is to replace purely ordinal confidence classes with empirically evaluated calibration where representative labels are available. Calibration should remain claim-specific: identity probability, exact-version probability, and affected-range confidence answer different questions. Set-valued version predictions are also promising for releases that are indistinguishable under surviving APK evidence. Their quality should be measured by coverage and set size rather than by rewarding arbitrary single-version guesses.

5.5.2 Independent and Human-Centered Evaluation

The framework should be tested with analysts who did not design it and with detectors that expose different evidence families. Such studies can measure whether reviewers assign consistent claim classes, trace outputs to the same provenance, recognize when vulnerability authority is capped, and make safer decisions than users of a binary present/absent report.

5.5.3 Interoperable Provenance

Reconstructed SBOM producers need a shared vocabulary for observed binary evidence, inferred identity, candidate version sets, external validation, and abstention. Future work can explore CycloneDX extensions or profiles that preserve these distinctions while remaining consumable by vulnerability management and policy tools. Cross-tool evidence exchange would also allow an analyst to combine independent signals without erasing their origin.

5.5.4 Broader Component Domains

Native, dynamic, and protected components should be treated as separate research domains rather than forced into Maven-oriented claims. Native attribution may use symbols, strings, sections, function fingerprints, build identifiers, and architecture-normalized features.

Runtime collection should record where a payload was obtained, when it was loaded, and whether it was packaged, unpacked, generated, or downloaded.

5.5.5 Evidence Knowledge Graphs

A provenance-aware knowledge graph could connect package coordinates, target variants, release families, reference artifacts, binary observations, source validation, experiments, and vulnerability ranges. The graph could explain why a candidate was considered and which evidence conflicts with it. Its relations must remain supporting evidence rather than a substitute for observed APK content.

5.6 Closing Remarks

A recovered SBOM is trustworthy only when its precision of language matches the precision of its evidence. Exact claims should be exact for a reason; ambiguous claims should remain ambiguous; unsupported regions should be visible; and vulnerability conclusions should stop where component evidence stops. The long-term objective is therefore not a system that always returns a complete dependency list. It is a system that can explain why each returned claim should be believed, what remains unknown, and how the result may safely guide action.

Appendix A

Literature Review Protocol

This appendix records version 1.0 of the structured literature mapping protocol, frozen on 21 June 2026. It is the operational companion to the methodology summarized in chapter 2. The review follows the planning, selection, quality-assessment, extraction, and synthesis principles of software-engineering review guidance [10], while explicitly acknowledging that the seed corpus was assembled iteratively before the protocol was frozen.

A.1 Research Questions

RQ-R1 Which evidence sources are used to identify Android third-party components from compiled applications?

RQ-R2 Which techniques support library identity, exact-version attribution, or both?

RQ-R3 Which compilation, optimization, obfuscation, packaging, and dependency transformations threaten each evidence source?

RQ-R4 How do studies construct ground truth and evaluate library-level, version-level, and vulnerability-related claims?

RQ-R5 How do studies represent confidence, ambiguity, provenance, abstention, partial inclusion, and conflicting evidence?

RQ-R6 Which research gaps prevent recovered component claims from becoming trustworthy and auditable SBOM statements?

A.2 Search Strategy

A.2.1 Sources and Scope

The primary search sources are ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, USENIX, and arXiv. Google Scholar is used for discovery and backward or forward citation chaining rather than as the sole bibliographic authority. The publication window begins in 2015 and ends on the date recorded for each search refresh. English full text is preferred; English title and abstract are mandatory for screening.

A.2.2 Concept Blocks

Table A.1: Search concept blocks

Block	Terms
Platform	Android, APK, DEX, Dalvik
Component	third-party library, TPL, software component, dependency, SBOM, software bill of materials
Task	detect*, identif*, recogn*, recover*, attribut*, fingerprint*, match*
Threat or specificity	obfuscat*, R8, ProGuard, optim*, shrink*, inlin*, version*, vulnerab*

The canonical query is:

```
(Android OR APK OR DEX OR Dalvik)
AND ("third-party library" OR TPL OR "software component" OR dependency
   OR SBOM OR "software bill of materials")
AND (detect* OR identif* OR recogn* OR recover* OR attribut*
   OR fingerprint* OR match*)
AND (obfuscat* OR R8 OR ProGuard OR optim* OR shrink* OR inlin*
   OR version* OR vulnerab*)
```

Database-specific syntax may change, but a refresh must preserve the four concept blocks and record the exact executed query, date, raw result count, deduplicated count, and selection decisions.

The frozen seed corpus predates this audit requirement. Its original database-specific query executions, raw result counts, and deduplicated flow are therefore not claimed as retrospectively available.

A.3 Inclusion and Exclusion Criteria

A.3.1 Inclusion

A publication is included when it:

1. addresses component, library, package, or version recovery from Android binaries or evaluates tools that do;
2. studies evidence obtainable from compiled artifacts, reference libraries, package metadata, or related dependency knowledge;
3. contributes a method, benchmark, empirical assessment, taxonomy, or SBOM-oriented framework relevant to compiled-artifact attribution;
4. permits extraction of at least the task, evidence source, evaluation basis, and limitations; and
5. has an accessible English title and abstract.

A.3.2 Exclusion

A publication is excluded when it:

1. relies only on source dependency manifests and does not study binary recovery;
2. concerns malware classification, clone detection, or vulnerability scanning without a component-attribution contribution;
3. is a tutorial, slide deck, blog post, product page, or duplicate;
4. reports no separable Android result;
5. lacks enough information to identify its method or evaluation basis; or
6. is a shorter version of an included extended study and contributes no distinct method, dataset, or result.

A.3.3 Screening and Evidence Tiers

Records are deduplicated and screened by title and abstract. Full-text eligibility screening and structured extraction are completed for Tier A. Records without a verified local full text remain Tier B pending that stage. The evidence tiers are:

Tier A — full-text verified A local full text has been inspected and the publication may support structured and quantitative extraction.

Tier B — supplementary record Metadata or a structured note exists, but the local full text has not yet been verified. The record may guide search and provisional gap mapping, but not precise quantitative claims.

Tier C — contextual authority Standards, official documentation, and project artifacts used for definitions or framework design. They are not counted as primary studies.

Excluded The reason for exclusion is recorded and the item is not used in synthesis.

A.4 Data Extraction Schema

Table A.2: Structured extraction fields

Dimension	Extracted information
Bibliographic identity	Title, authors, year, venue, DOI or stable URL, publication type, study family, evidence tier.
Objective and task	Research objective; family, library, class, version, SBOM, or vulnerability granularity.
Inputs and assumptions	Required APK artifacts, reference corpus, metadata, training data, and coverage assumptions.
Segmentation	Package boundaries, class hierarchy, dependency graph, clustering, or other module-decoupling method.

Dimension	Extracted information
Evidence	Names, APIs, opcodes, control flow, call/dependency graphs, literals, resources, native code, metadata, learned representations, and cross-component relations.
Decision method	Hashing, similarity, graph matching, ranking, classification, probability, learned model, or evidence fusion.
Uncertainty behavior	Threshold, confidence, competing versions, abstention, partial matching, provenance, and conflict handling.
Transformations	Renaming, package flattening, shrinking, optimization, control-flow rewriting, method inlining, repackaging, multidex, partial inclusion, native code, and dynamic loading.
Evaluation	Dataset, ground truth, evaluation unit, metrics, baselines, obfuscation setting, and independent benchmark results.
Operational support	Artifact availability, runtime, scalability, SBOM format, vulnerability mapping, and transitive-dependency handling.
Limitations and relevance	Threats to validity, negative findings, and adoption, adaptation, or rejection relevance to the framework.

Unreported information is recorded as `NR`; it is not inferred from neighboring fields or from another publication in the study family.

A.5 Quality Assessment

Tier A publications receive a score of 1, 0.5, or 0 for each question:

1. Is the research objective clear?
2. Is the method described well enough to understand or reproduce?
3. Are the dataset and ground truth described?
4. Are the task unit and metrics explicit?
5. Are transformation or obfuscation conditions stated?
6. Are identity and exact-version claims separated?
7. Is an implementation, dataset, or artifact available?
8. Are limitations or threats to validity acknowledged?

The score controls synthesis weight rather than automatic inclusion. Historically important studies may remain in the map with an explicit reporting-quality limitation.

A.5.1 Completed Quality Assessment

Table A.3 records the completed assessment for the 15 current Tier A studies. The vector lists Q1–Q8 in the order defined above; totals are out of eight. Scores reflect reporting and reproducibility in the inspected publication, not the scientific importance of the work.

Table A.3: Reporting-quality assessment for current Tier A studies

ID	Study	Q1–Q8 score vector	Total
S01	LibScout	1, 1, 1, 1, 1, 1, 1, 1	8.0
S02	LibD	1, 1, 0.5, 1, 0.5, 0.5, 0.5, 1	6.0
S03	LibPecker	1, 1, 1, 1, 1, 0.5, 0.5, 1	7.0
S04	ORLIS	1, 1, 1, 1, 1, 0.5, 0.5, 1	7.0
S05	LibID	1, 1, 1, 1, 1, 1, 0.5, 1	7.5
S06	ASE 2020 assessment	1, 1, 1, 1, 1, 1, 1, 1	8.0
S07	Systematic assessment	1, 1, 1, 1, 1, 1, 0.5, 1	7.5
S08	ATVHUNTER	1, 1, 1, 1, 1, 1, 0.5, 1	7.5
S09	LibScan	1, 1, 1, 1, 1, 1, 0.5, 1	7.5
S10	Ramirez thesis	1, 1, 1, 1, 0.5, 1, 0.5, 1	7.0
S11	LibHunter	1, 1, 1, 1, 1, 1, 0.5, 1	7.5
S12	LibAttention	1, 1, 1, 1, 1, 1, 0.5, 1	7.5
S13	SAD	1, 1, 1, 1, 1, 1, 0.5, 1	7.5
S17	TPL-Benchmark	1, 1, 1, 1, 1, 1, 1, 1	8.0
S18	UniBOM	1, 1, 0.5, 0.5, 0.5, 0.5, 1, 0.5	5.5

A.6 Synthesis Rules

- Narrative and taxonomy-based synthesis is the default.
- Incompatible datasets and metrics are not pooled into a common leaderboard.
- Library-level, class-level, and version-level results remain separate.
- Original-paper results remain separate from independent benchmark results.
- Obfuscation and optimization conditions must accompany performance claims.
- Artifact availability is assessed independently of reported accuracy.
- Tier B records cannot independently support precise numerical claims.
- Identity authority, version authority, and vulnerability authority are separate.
- Explicit ambiguity or abstention is preferred to unsupported exactness.

A.7 Selected Studies

Table A.4: Frozen seed corpus and current evidence status

ID	Publication or study label	Year	Status
S01	LibScout [13]	2016	Tier A
S02	LibD [16]	2017	Tier A
S03	LibPecker [17]	2018	Tier A
S04	ORLIS [14]	2018	Tier A
S05	LibID [15]	2019	Tier A
S06	Automated Third-Party Library Detection: Are We There Yet? [11]	2020	Tier A
S07	Systematic Assessment of Android TPL Detection Tools [12]	2021	Tier A

ID	Publication or study label	Year	Status
S08	ATVHUNTER [18]	2021	Tier A
S09	LibScan [19]	2023	Tier A
S10	Empirical Study of TPL Detection in iOS and Android [23]	2024	Tier A
S11	LibHunter / Code-Optimization Impact Study [20]	2024	Tier A
S12	LibAttention [22]	2025	Tier A
S13	SAD [21]	2025	Tier A
S14	LibRadar	2016	Tier B
S15	Libra	2023	Tier B
S16	LibMD	2025	Tier B
S17	TPL-Benchmark / Revisiting TPL Detection [24]	2025	Tier A
S18	UniBOM [25]	2025	Tier A
S19	LibScope / Towards Global Matches	2026	Tier B
S20	ObfDetector / Advanced-Obfuscation Robustness Study	2026	Tier B

The seed corpus contains 20 publication units: 15 Tier A full texts and five Tier B supplementary records. This count is a protocol checkpoint, not a claim that all 20 have equal evidential authority. A Tier B record is promoted only after full-text verification, bibliographic checking, structured extraction, and quality assessment.

A.8 Audit and Change Control

The current extraction records are maintained under `papers/data/`. Each search refresh must append the date, sources, query variants, raw and deduplicated result counts, inclusion and exclusion decisions, tier changes, and taxonomy effects to the master-thesis session log. Eligibility, evidence tiers, and synthesis rules may be amended only through a dated protocol change note; silent changes are prohibited.

A.9 Protocol Change Notes

A.9.1 Version 1.0a — Screening Clarification

The full-text screening stage is complete only for records with a locally verified full text. The frozen seed corpus predates the protocol, so its original database-specific query counts cannot be reconstructed as an exhaustive selection flow.

A.9.2 Version 1.0b — Full-Text Promotions

On 21 June 2026, TPL-Benchmark and UniBOM were inspected locally, checked bibliographically, extracted under the protocol schema, and quality-scored. They were promoted from Tier B to Tier A. TPL-Benchmark supplies direct Android benchmark evidence;

UniBOM supplies adjacent binary/SBOM workflow evidence and is not used to support Android Maven exact-version claims.

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